

# Cambridge IGCSE™

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**DESIGN AND TECHNOLOGY****0445/53**

Paper 5 Graphic Products

**May/June 2026**

MARK SCHEME

Maximum Mark: 50

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**Published**

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

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This document consists of **9** printed pages.

**General marking principles**

All examiners must apply these marking principles when marking candidate responses.

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Examiners must award marks for each response in line with: <ul style="list-style-type: none"> <li>the specific <b>content</b> defined in the mark scheme</li> <li>the specific <b>skills</b> defined in the mark scheme or in the level descriptions</li> <li>the <b>standard of response</b> required as exemplified by the standardisation scripts.</li> </ul>                                                                                                                                                                                                                                                                                    |
| 2 | Examiners must award <b>whole marks</b> (not half marks, or other fractions).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3 | Examiners must award marks <b>positively</b> . This means that: <ul style="list-style-type: none"> <li>marks are not deducted for errors or omissions</li> <li>marks are awarded for correct/valid answers as defined in the mark scheme and also for <b>other valid answers</b> which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate</li> <li>the quality of spelling, punctuation and grammar are judged <b>only</b> when the mark scheme indicates that these features are <b>specifically assessed</b> in the question. However, the meaning of all responses must be unambiguous.</li> </ul> |
| 4 | Examiners must <b>consistently</b> apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 5 | Examiners must use the <b>full range of marks</b> defined in the mark scheme, unless limited by the quality of the candidate responses seen.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 6 | Examiners must award marks according to the mark scheme and <b>must not</b> award marks with grade thresholds or grade descriptions in mind.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

**Annotations guidance for centres**

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

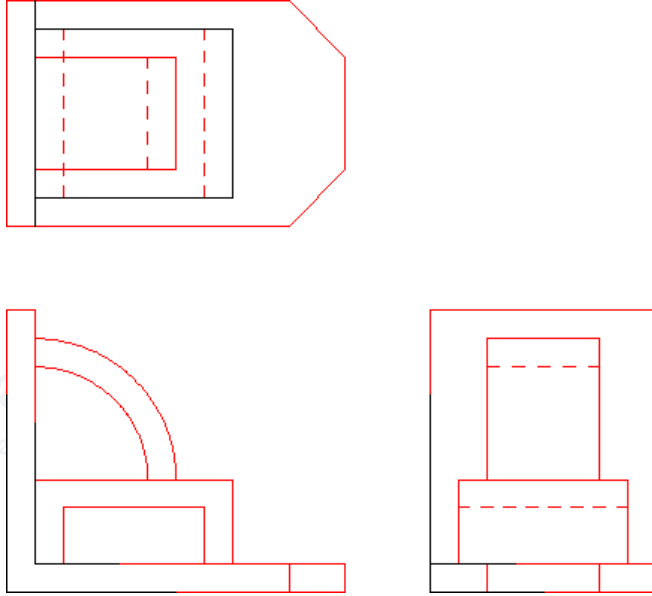
**Annotations**

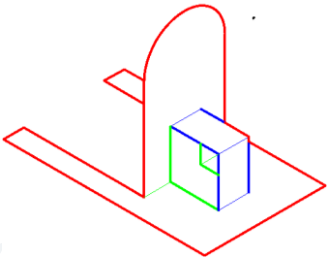
| Annotation                                                                        | Meaning                                                               |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------|
|  | Incorrect point                                                       |
|  | Indicates that the point has been noted, but no credit has been given |
|  | Correct point                                                         |
| Numbers                                                                           | Indicating the mark allocated for the response                        |

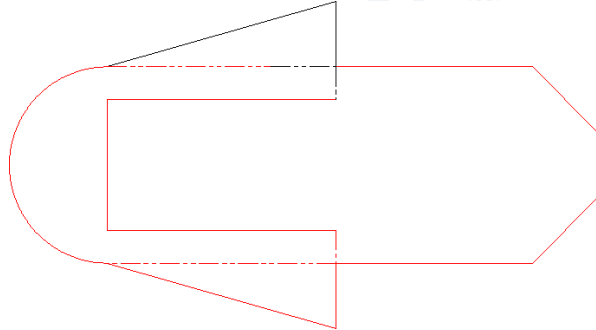
**PUBLISHED**

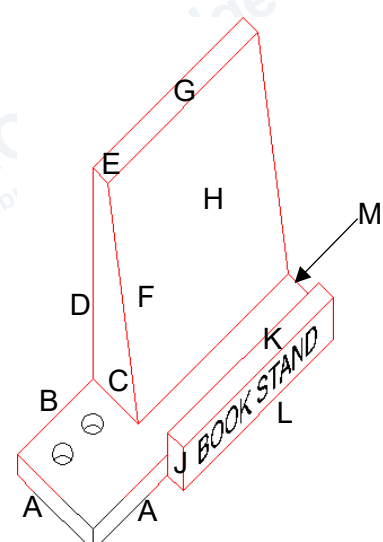
| Question   | Answer                                                                                                                                                                                                                                                                                                       | Marks | Guidance |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------|
| A1(a)(i)   | Circle to overlay [1]                                                                                                                                                                                                                                                                                        | 1     |          |
| A1(a)(ii)  | Left side bottom line mirror image of given right side line [1]<br>Top two lines correct 30° angle to overlay [1]<br>Top arc correct to overlay / candidate solution from centre of head circle [1]                                                                                                          | 3     |          |
| A1(a)(iii) | Vertical centre line added correct to overlay [1]<br><br>Right side base line and vertical edge correct to overlay [1]<br>Left side top angled line and vertical line correct to overlay [1]<br>Left side bottom angled line and vertical correct to overlay / candidate solution (mirror of right side) [1] | 4     |          |
| A1(a)(iv)  | Any octagon (8 sided shape drawn) [1]<br>Any regular octagon [1]<br>Octagon drawn correct to overlay [1]                                                                                                                                                                                                     | 3     |          |
| A1(b)(i)   | Square 30 mm high or 30 mm wide [1]<br>Square correct to overlay [1]                                                                                                                                                                                                                                         | 2     |          |
| A1(b)(ii)  | Any isosceles triangle [1]<br>Isosceles triangle correct to overlay [1]                                                                                                                                                                                                                                      | 2     |          |
| A2         | Outer rectangle 220 wide [1]<br>Outer rectangle 150 high [1]<br>25 mm radius to all four corners [1]<br>Even space at left and right [1]                                                                                                                                                                     | 4     |          |
| A3(a)      | Design shows some connection to talking or not talking [1]<br>No words used [1]<br>High quality sketch that conveys message clearly [1]                                                                                                                                                                      | 3     |          |

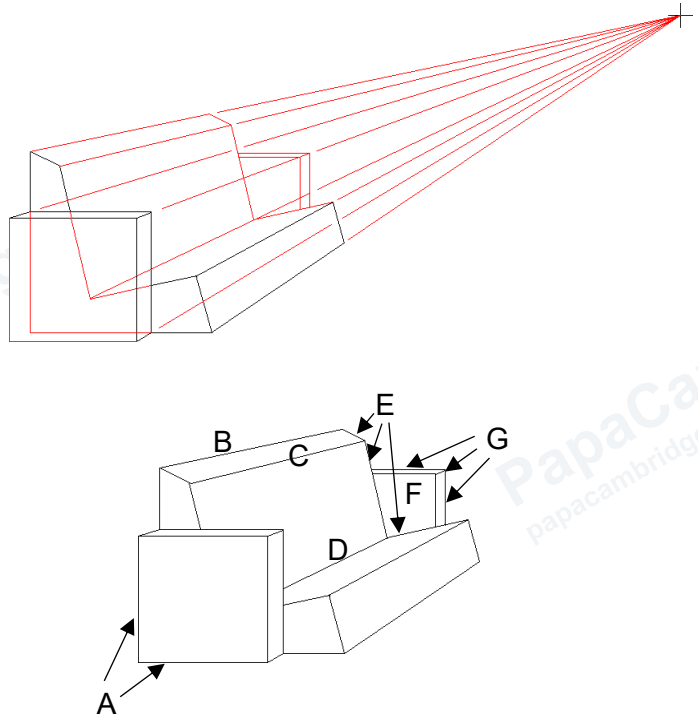
## PUBLISHED

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks | Guidance                                                                            |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------|
| A3(b)    | Outer circle drawn correct to overlay [1]<br>Diagonal bar added at 45° [1]<br>Diagonal bar correct to overlay [1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3     |                                                                                     |
| B4(a)    | <p>Side view:</p> <p>Vertical and horizontal sides of 'L' shape outline correct to overlay [1]<br/>Angled end section (vertical line) correct to overlay [1]<br/>Top and right side of U section added correct to overlay [1]</p> <p>3 inner faces of U section added 5mm thickness [1]<br/>Curved section correct (both curved lines) to overlay [1]</p> <p>Plan:</p> <p>Outline of base and back section projected from side view [1]<br/>Outline of curved section projected from side view [1]<br/>All 3 hidden detail lines added correctly [1]</p> <p>Front view:</p> <p>Vertical back outer rectangle correct to candidate solution [1]<br/>Front of base including angled sections correct to candidate solution [1]</p> <p>Curved section correct to candidate solution [1]<br/>Bottom U section correct to candidate solution [1]<br/>Hidden detail to curved section and U - section correct [1]</p> | 13    |  |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                     | Marks | Guidance                                                                            |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------|
| B4(b)(i) | <p>Outline of whole shape and arch in thick line <b>RED LINES</b> [1]</p> <p>Top two edges 'A' and right two edges 'B' of pushed out section thick and three folds in thin <b>BLUE LINES</b> [1]</p> <p>Two left sides and two bottom edges 'C' of pushed out section thick and two bottom crease lines to base of upright thin <b>GREEN LINES</b> [1]</p> | 3     |  |

| Question  | Answer                                                                                                                                                                                                                                                                               | Marks | Guidance                                                                                                                                                                                                                           |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B4(b)(ii) | File / wet and dry paper [1]<br>Line bender / strip heater / hot air / heat gun [1]                                                                                                                                                                                                  | 2     | Do not accept sandpaper                                                                                                                                                                                                            |
| B4(c)     | 30mm radius curve correct to overlay [1]<br>Inner 3 sides of left section correct to overlay [1]<br>Right hand section correct to overlay [1]<br>Triangular side support at bottom correct to overlay [1]<br>Four fold lines added to correct convention (2 dots & dash) [1]         | 5     |                                                                                                                                                 |
| B4(d)     | More accurate [1] as computer can measure to 1000ths of a mm [1]<br>Quicker than drawing by hand [1] as you can copy and paste parts [1]<br>Quicker to send [1] as can be emailed [1]<br>Easy to edit / revise [1] as parts can be deleted / added without re-drawing [1]<br>Or AOVR | 2     | 1 mark for naming a benefit, second mark for description.<br>Award two marks for description or 1 mark for two benefits stated.<br>Do not accept 'easier' / 'quicker' as a benefit unless qualified e.g. Easier to amend / revise. |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks | Guidance                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------|
| B5(a)    | <p>Front and end faces of left side 'A' correct to overlay [1]<br/>           Back edge 'B' correct length and positioned to candidate solution [1]<br/>           Side edge 'C' correct length and positioned to candidate solution [1]<br/>           Side 'D' correct length and positioned to candidate solution [1]<br/>           Top edge 'E' correct length and positioned to candidate solution [1]<br/>           Sloping edge 'F' correct to overlay / candidate solution [1]</p> <p>Top face 'G' correct length and angle [1]<br/>           Front face 'H' correct to overlay / candidate solution [1]</p> <p>End face 'J' correct size to overlay [1]<br/>           Top face 'K' correct length [1]<br/>           Front face 'L' correct to overlay / candidate solution [1]<br/>           End line 'M' correct to overlay / candidate solution [1]</p> | 12    |                                                |
| B5(b)    | <p>Appropriate colour (brown / orange / yellow) added [1]<br/>           Grain lines added [1]<br/>           High quality rendering – end grain shown and block looks like wood [1]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3     |                                                                                                                                   |
| B5(c)(i) | <p>To check proportions / sizes / function [1] of product before manufacture [1]</p> <p>To prevent wasting materials [1] which can be expensive [1]<br/>           To test ideas [1] quickly and easily [1]<br/>           Or AOVR</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     | <p>1 mark for naming a reason, second mark for description. Award two marks for description or 1 mark for two reasons stated.</p> |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B5(c)(ii) | Lightweight / easy to cut, shape and join / inexpensive [1]                                                                                                                                                                                                                                                                                                                                                                                                                           | 1     | Do not allow: availability<br>Accept any other valid response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| B5(d)     | <p>Horizontal and vertical front end lines 'A' correct to overlay (armrest) [1]</p> <p>Top back line 'B' to VP [1]</p> <p>Lower back line 'C' to VP [1]</p> <p>Lower back edge 'D' in correct position and to VP [1]</p> <p>3 lines 'E' for back edge of centre piece parallel to front lines correct to overlay [1]</p> <p>Front face of right side end piece 'F' on correct projection lines [1]</p> <p>Back edges and corner 'G' added in proportion to candidate solution [1]</p> | 7     |  <p>The diagram shows two views of a chair backrest. The top view is a perspective drawing with red construction lines converging to a vanishing point on the right. The bottom view is an elevation drawing with labels A through G. Label A points to the front edge of the seat. Label B points to the top back edge. Label C points to the lower back edge. Label D points to the lower back edge of the seat. Label E points to the back edge of the central piece. Label F points to the front face of the right side end piece. Label G points to the back edge and corner of the right side end piece.</p> |